EXAMPLE 2.1

A particle moves along a straight line such that its position is given by: $s = 2t^3 - 8t^2 + 10t + 20\,m$ where t is the time in seconds. Determine the velocity and the acceleration of the particle when t = 4 seconds. What will be the displacement in the first 10 seconds.

SOLUTION

$s = 2t^3 - 8t^2 + 10t + 20\,m$

$v = \frac{ds}{dt}$

$v = 6t^2 - 16t + 10\,m/s$

$a = \frac{dv}{dt}$

$a = 12t - 16\,m/s^2$

When t = 4 seconds:

$v = 42\,m/s$

$a = 32\,m/s^2$

To determine the displacement in the first 10 seconds:

$s_0 = 20m$

$s_{10} = 2 \times 10^3 - 8 \times 10^2 + 10 \times 10 + 20 = 1320\,m$

$\Delta s = s_{10} - s_0 = 1300m$

EXAMPLE **2.2**

A particle is moving with an acceleration given by $a = 12t^2\ m/s^2$. When t = 0, s = 0 and, the velocity of the particle is 20 m/s. What is the particle velocity when t = 2 seconds. What is the position of the particle when t = 10 seconds.

SOLUTION

$$\int_{20}^{v} dv = \int_{5}^{t} adt$$

$$[v]_{20}^{v} = \int_{5}^{t} 12t^2 dt = [4t^3]_0^t$$

$$v = 4t^3 + 20\ m/s$$

When t = 2 s

$$v = 52\ m/s$$

$$\int_{0}^{s} ds = \int_{0}^{10} vdt$$

$$s = \int_{0}^{10} [4t^3 + 20]dt = [t^4 + 20t]_0^{10} = 10200m$$

EXAMPLE 2.3

The car on the left in the photo and in Fig. moves in a straight line such that for a short time its velocity is defined by $v = (3t^2 + 2t)$ m/s, where t is in seconds. Determine its position and acceleration when $t = 3$ s. When $t = 0$, $s = 0$.

Fig.

SOLUTION

Coordinate System. The position coordinate extends from the fixed origin O to the car, positive to the right.

Position. Since $v = f(t)$, the car's position can be determined from $v = ds/dt$, since this equation relates v, s, and t. Noting that $s = 0$ when $t = 0$, we have*

$$(\overset{+}{\rightarrow}) \qquad v = \frac{ds}{dt} = (3t^2 + 2t)$$

$$\int_0^s ds = \int_0^t (3t^2 + 2t)dt$$

$$s\Big|_0^s = t^3 + t^2\Big|_0^t$$

$$s = t^3 + t^2$$

When $t = 3$ s,

$$s = (3)^3 + (3)^2 = 36 \text{ m} \qquad \textit{Ans.}$$

Acceleration. Since $v = f(t)$, the acceleration is determined from $a = dv/dt$, since this equation relates a, v, and t.

$$(\overset{+}{\rightarrow}) \qquad a = \frac{dv}{dt} = \frac{d}{dt}(3t^2 + 2t)$$

$$= 6t + 2$$

When $t = 3$ s,

$$a = 6(3) + 2 = 20 \text{ m/s}^2 \rightarrow \qquad \textit{Ans.}$$

NOTE: The formulas for constant acceleration *cannot* be used to solve this problem, because the acceleration is a function of time.

*The *same result* can be obtained by evaluating a constant of integration C rather than using definite limits on the integral. For example, integrating $ds = (3t^2 + 2t)dt$ yields $s = t^3 + t^2 + C$. Using the condition that at $t = 0$, $s = 0$, then $C = 0$.

EXAMPLE 2.4

A particle moves along a straight line with an acceleration given by: $a = -4s\ m/s^2$. Where s is the position in meters. If the initial conditions are, when t = 0, s = 0 and v = 10 m/s. Determine the velocity of the particle when s = 4 m. Determine also the position at which the particle momentarily stops.

SOLUTION

$$a = v\frac{dv}{ds} = -4s$$

$$vdv = -4sds$$

$$\int_{10}^{v} vdv = -4\int_{0}^{s} sds$$

$$v^2 = 100 - 4s^2$$

When s = 4m:

$$v^2 = 100 - 4 \times (4)^2 = 36$$

$$v = 6\ m/s$$

When v = 0:

$$100 - 4s^2 = 0$$

$$\therefore s = 5m$$

EXAMPLE 2.6

A car moves along a straight line with a velocity of 90 km/hr. If the driver uses the brakes to stop the car after a time of 2 seconds, determine the constant deceleration and the distance the car moves before it stops.

SOLUTION

The initial velocity $\boldsymbol{v_0 = 90km/hr = 90 \times \frac{5}{18} = 25m/s}$, the final velocity v = 0, the time t = 2 sec. and the acceleration is constant.

$$v = v_0 + at$$

$$0 = 25 + 2a$$

$$a = -12.5m/s^2$$

$$s = v_0 t + \frac{1}{2}at^2$$

$$s = 25 \times 2 + \frac{1}{2}(-12.5) \times (2)^2 = 25m$$

PROBLEMS

2-1 A particle is moving along a straight line such that its position is given by $s = t^3 - 6t^2 + 9t$ m, where t is in seconds. Determine the displacement, velocity, and the acceleration after 2 seconds.

2-2 A particle is moving in straight line with an acceleration given by $a = 4t^3 - 3\ m/s^2$ where t is the time in seconds. Initially when t = 0, v = 5 m/s and s = 0. Determine the particle's velocity and position when t = 4 s.

2-3 A particle is moving in straight line with a velocity $v = 4t^3 - 2t\ m/s$. where t is the time in seconds. Initially when t = 0 the position s = 2 m, determine the particle's position and acceleration when t = 5 s.

2-4 A particle travels along a straight line with an acceleration of $a = 10 - 0.2s\ m/s^2$, where s is measured in meters. Determine the velocity of the particle when s = 10 m if v = 5 m/s at s = 0.

2-5 A particle is moving in a straight line path with an acceleration given by $a = 3v^{-2}\ m/s^2$, where v is the velocity in m/s. Determine the particle's velocity when s = 10 m. What is the particle position when v = 15m/s.

2-6 Initially, the car travels along a straight road with a speed of 35 m/s. If the brakes are applied and the speed of the car is reduced to 10 m/s in 15 s, determine the constant deceleration of the car.